

Practice 4 - Methods of organizing fail-safe communication channels

Background

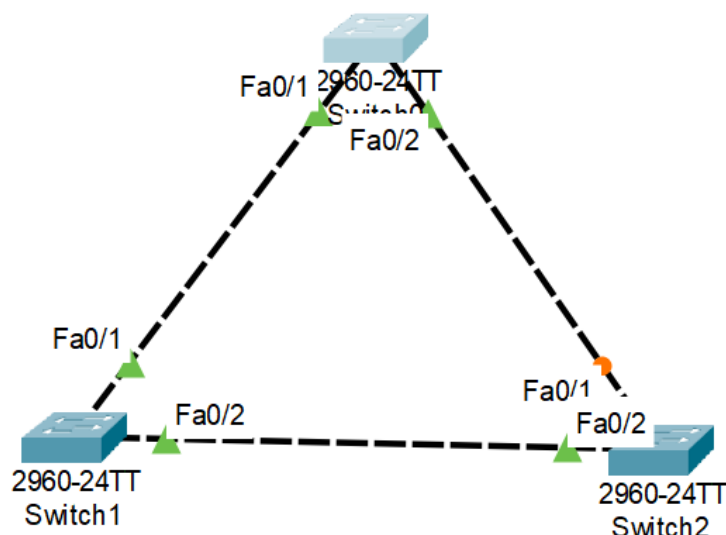
In this lab, we will discuss loop prevention using the Spanning Tree Protocol (STP). STP is a network protocol designed to prevent switching loops in redundant network topologies by selectively blocking redundant links while ensuring network stability.

There are two main methods of fault tolerance: **link redundancy** (where one link is active, and the other is on standby) and **link aggregation** (which combines multiple physical links into one logical connection for increased bandwidth and resilience).

Part 1: Link Redundancy

1.1 Exploring STP set up

Step 1: Build the network using the topology shown below.



Step 2: Configure Basic STP on Switches.

1. Access the CLI of each switch and enter privileged mode:
`Switch>enable`
2. Verify STP settings:
`Switch#show spanning-tree`
3. Identify which switch is the **Root Bridge** and why.

4. Check port roles: **Root Port (RP)**, **Designated Port (DP)**, and **Blocked Port**. Which port is in a blocking state (BLK) and why?
5. What other state do you know?

The image shows three screenshots of Cisco IOS Command Line Interface (CLI) for different switches, illustrating STP (Spanning Tree Protocol) configuration and status.

Switch0 CLI:

```
Switch>
Switch>en
Switch#sho sp
Switch#sho spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
            Address     0001.9746.BD6B
            Cost       19
            Port       1(FastEthernet0/1)
            Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
  Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
            Address     000B.BEB1.BC4E
            Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
            Aging Time 20
Interface Role Sts Cost Prio.Nbr Type
-----
Fa0/1     Root FWD 19   128.1 P2p
Fa0/2     Desg FWD 19   128.2 P2p
Switch#
```

Switch2 CLI:

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/2, changed state to up

Switch>en
Switch#sh sp
Switch#sh spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
            Address     0001.9746.BD6B
            Cost       19
            Port       2(FastEthernet0/2)
            Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
  Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
            Address     00D0.BABA.46A6
            Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
            Aging Time 20
Interface Role Sts Cost Prio.Nbr Type
-----
Fa0/1     Altn BLK 19   128.1 P2p
Fa0/2     Root FWD 19   128.2 P2p
Switch#
```

Switch1 CLI:

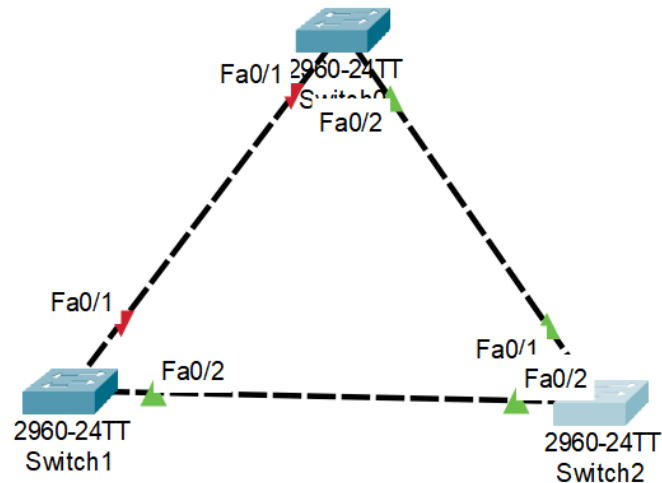
```
%LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/2, changed state to up

Switch>en
Switch#sh sp
Switch#sh spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
            Address     0001.9746.BD6B
            Cost       19
            Port       2(FastEthernet0/2)
            Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
  Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
            Address     0001.9746.BD6B
            Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
            Aging Time 20
Interface Role Sts Cost Prio.Nbr Type
-----
Fa0/1     Desg FWD 19   128.1 P2p
Fa0/2     Desg FWD 19   128.2 P2p
Switch#
```

Step 3: Simulate STP operation by blocking a redundant link:

1. Identify a port in **blocking mode** using `show spanning-tree`. In our example, it is the **Switch2 Fa0/1 port**.
2. **Disable an active link** (in our example Switch0 Fa0/1) and observe STP convergence (30-50 seconds):
 - a. Access Switch0 CLI and go to global configuration mode:
Switch#**configure terminal**
 - b. Shutdown a port with an active link:
Switch(config)#**interface FastEthernet 0/1**
Switch(config-if)#**shutdown**

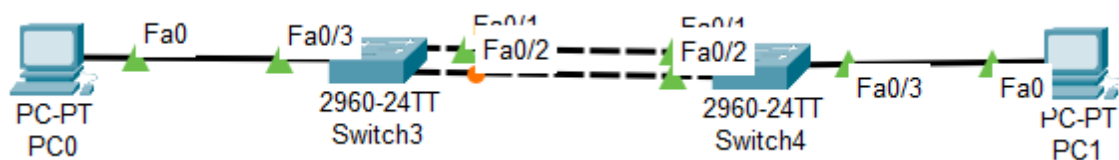
3. Check if the blocked port becomes active (forwarding mode). Access Switch2 CLI and check STP configuration:
Switch#**show spanning-tree**



1.2 Configuring RSTP (Rapid STP) for Faster Convergence

Step 1: Build the Topology

1. Open Cisco Packet Tracer.
2. Add the following devices to the workspace:
 - **2 Cisco 2960 switches** (Switch3, Switch4).
 - **2 PCs** (PC0, PC1).



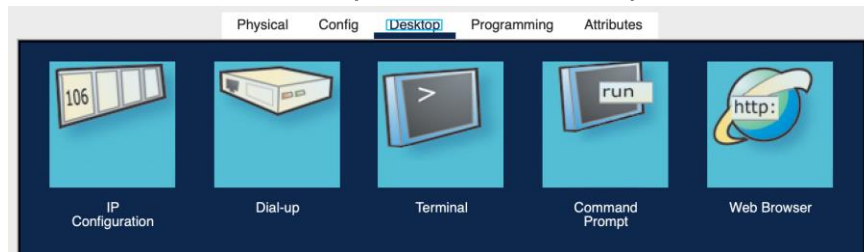
Step 2: Assign IP Addresses to PCs

1. **On PC0:**
 - IP Address: **192.168.1.1**
 - Subnet Mask: **255.255.255.0**
2. **On PC1:**

- IP Address: 192.168.1.2
- Subnet Mask: 255.255.255.0

3. Test connectivity using a ping from PC0 to PC1:

- Access Command Prompt of PC0 in Desktop tab:



- Enter command:
ping 192.168.1.2

Step 3: Enable RSTP on Switches

1. Access the CLI of each switch and enter global configuration mode:
Switch>**enable**
Switch#**configure terminal**
2. Enable RSTP mode:
Switch(config)#**spanning-tree mode rapid-pvst**
3. Verify RSTP settings:
Switch#**show spanning-tree**
4. Identify which switch is the **Root Bridge**.
5. Check port roles: **Root Port (RP)**, **Designated Port (DP)**, and **Blocked Port**.

Step 4: Test STP Convergence Time

1. Disable an active link (Simulate Failure) on Switch4:

```
Switch(config)#interface FastEthernet 0/1
Switch(config-if)#shutdown
```

Observe network behavior: The blocked port transitions through states (**Listening** → **Learning** → **Forwarding**).

2. Test connectivity again during failover:

```
ping 192.168.1.2
```

With standard STP, the failover takes **30-50 seconds**, while with RSTP, the failover should be **instant (sub-second switching)**.

3. Reactivate the disabled link. Enter Switch4 CLI:
`Switch(config)#interface FastEthernet 0/1`
`Switch(config-if)#no shutdown`

The network should return to normal immediately.

4. Use `show spanning-tree` to check **active, root, and blocked ports**.
5. Test connectivity between PCs again.

Part 2: Link Aggregation

2.1 Static aggregation

Step 1: Build the Topology

1. Open Cisco Packet Tracer.
2. Add the following devices to the workspace:
 - **2 Cisco 2960 switches** (Switch0, Switch1).
 - **2 PCs** (PC0, PC1).



Reminder: Connect the PC to fa0/3 switch's port!

Step 2: Assign IP Addresses to PCs

On PC0:

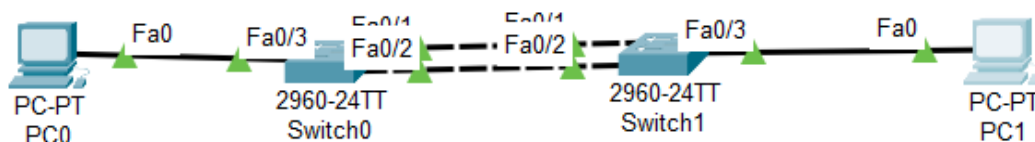
- IP Address: `192.168.1.1`
- Subnet Mask: `255.255.255.0`

On PC1:

- IP Address: `192.168.1.2`
- Subnet Mask: `255.255.255.0`

Step 3: Configure Static EtherChannel

1. Access the CLI of each switch and enter global configuration mode:
Switch>**enable**
Switch#**configure terminal**
2. Select interfaces for aggregation on Switch0 and Switch1:
Switch(config)#**interface range FastEthernet 0/1-2**
3. Create the EtherChannel (Static Mode):
Switch(config-if-range)#**channel-group 1 mode on**
4. Save the configuration:
Switch(config-if-range)#**end**
Switch#**write memory**
5. Repeat the same steps on Switch1.
6. Verify the configuration:
Switch#**show etherchannel summary**
7. Connect switches fa0/1-2 respectively:



***If there are different VLANs, you can configure a trunk port between switches as follows**

```
Switch#configure terminal
Switch(config)#interface port-channel 1
Switch(config)#switchport mode trunk
Switch(config)#switchport trunk allowed vlan xx
```

Part 3: Independent Task

Task: Network Analysis and Answering Key Questions

After completing all the steps of the lab work and configuring **RSTP** and **EtherChannel**, you need to analyze the network and answer the following questions.

Steps to Complete:

1. **Analyze the configured network** using the following commands:
 - show spanning-tree (identify the Root Bridge, port states, and blocked ports).
 - show etherchannel summary (check if EtherChannel is functioning properly).
2. **Disable one of the links** used by STP (shutdown on an interface) and observe the changes in traffic flow.
3. **Evaluate network performance** by testing connectivity between PCs (ping 192.168.1.2).
4. **Answer the key questions and prepare a report.**

Key Questions:

1. Which switch became the Root Bridge, and why?
2. Which ports were blocked by STP?
3. How did the traffic path change after disabling a link?
4. What are the advantages of using EtherChannel in this network?